

Calceus Serial Protocol v1

Serial Details

Communication details: using Nordic UART Service (BLE), with the application as controller and the insole as peripheral. The insole will be named either REIDLHS or REIDRHS, with insoles being identified by their MAC address.

RX Characteristic (UUID: 6E400002-B5A3-F393-E0A9-E50E24DCCA9E)

The controller can send data to the peripheral by writing to the RX Characteristic of the service. ATT Write Request or ATT Write Command can be used.

TX Characteristic (UUID: 6E400003-B5A3-F393-E0A9-E50E24DCCA9E)

If the controller has enabled notifications for the TX Characteristic, the peripheral can send data to the peer as notifications.

Packet Format

;(type)(subtype)[space][data...](line end)

; - signifies the start of packet. The packet decoder should ignore all data until this is received.

(type) - can be:

Q - Query (sent by the controller asking for data from the peripheral/device)

R - Response (sent by the peripheral/device in response to a query from the controller)

C - Command (sent by the controller asking for the peripheral/device to perform a function)

A - Acknowledge (sent by the peripheral/device in response to a command from the controller)

E - Error (sent by either device to indicate that there was a problem with the last message)

(subtype) - depends on whether it is a query or command. Specifies the type of command or query requested (if sent from the central/app), the type of command or query that was actioned (if sent from the peripheral/device), or the type of error that was detected.

[space][data] - optional field, where a space and then data may be appended if required by the message type. For example, a command to reboot the device does not require data, but a command to set a parameter to a certain value will require the value to be appended.

(line end) - this signifies the end of the packet. The packet decoder should accept either \r or \n for the line end character. Subsequent line end characters will be ignored since they are not the packet start character. When transmitting a packet, it is recommended that both are sent.

Error Types

Error Type	Character	Example	Description
Bad Command	C	;EC\r\n	Sent by a peripheral/device to indicate that the received command packet was either of an unsupported command type, or that the additional data required for the command was not supplied or in an invalid format.
Bad Query	Q	;EQ\r\n	Sent by a peripheral/device to indicate that the received query packet was either of an unsupported query type, or that the additional data required for the query was not supplied or in an invalid format.
Protocol Error	M	;EM\r\n	Sent by either a controller or a peripheral/device to indicate that there was an error with the packet structure, for example a packet end was not received within a reasonable time of the packet start, or far too many characters were received between packet start and packet end.
No Data	N	;EN\r\n	Sent by a peripheral in response to a query or command operating on data to indicate that the requested data could not be found in memory.
Unknown Error	U	;EU\r\n	Sent by either a controller or a peripheral/device to indicate that there was an error that was unable to be classified as one of the previous errors, for example if there was a baud rate mismatch.

Query/Response Types

Query Type	Character	Example	Description
System Info	I	;QI\r\n ;RI UID=13579BDF, VER=00010002	Returns the serial number and firmware version of the device in an ASCII string for easy debugging.
Battery Level	B	;QL\r\n ;RL 2975\r\n	Returns the measured battery voltage.
Time	T	;QT\r\n ;RT 1648097508\r\n	Returns the device time in seconds (seconds since boot if not already configured, otherwise it can be configured as an epoch).
All record indices	A	;QA\r\n ;RA 35,102\r\n	Returns the range of indexes currently in memory (each new record i.e. summary gets an incremental index number).
Pending record indices	P	;QP\r\n ;RP 97,102\r\n	Returns the range of indexes currently in memory that aren't marked as synced.
Record (single)	R	;QR 97\r\n ;RR [binary blob]\r\n	Returns the record with the specified index.

The binary blob has the following format:

```
#pragma pack(push,1)
typedef struct reid_ble_summary_packet_t {
    uint32_t time_s_start;
    uint32_t time_s_end;
    uint32_t index;
    uint16_t is_synced;
    uint16_t num_samples;
    uint16_t vdd_mv_min;
    uint16_t fsr1_max;
    uint16_t fsr2_max;
    uint16_t fsr3_max;
    uint16_t fsr4_max;
    uint16_t fsr5_max;
    uint16_t fsr6_max;
    uint16_t fsr7_max;
    uint16_t fsr8_max;
    uint16_t fsr9_max;
    uint16_t fsr10_max;
    int16_t ntc1_max;
    int16_t ntc2_max;
    int16_t ntc3_max;
    int16_t ntc4_max;
    int16_t ntc5_max;
    int16_t cap1_delta_min;
    int16_t cap1_delta_max;
    int16_t cap2_delta_min;
    int16_t cap2_delta_max;
    int16_t cap3_delta_min;
    int16_t cap3_delta_max;
    int16_t cap4_delta_min;
    int16_t cap4_delta_max;
} reid_ble_summary_packet_t;
#pragma pack(pop)
```

Query Type	Character	Example	Description
Multiple records	M	;QM 97,102\r\n	Returns all records with indices in the specified range, up to 100 at a time. Note that this could take a short while to transfer the data.
		;RM [binary blob]\r\n	Note that all data within the specified range will be returned, not just the data marked as synced.
		;RM [binary blob]\r\n	
		;RM [binary blob]\r\n	
		...	
		...	
		...	
		;RM [binary blob]\r\n	
Last data	L	;QL\r\n	Returns the most recent single measurement performed. This is useful for live data viewing and debugging.
		;RL [binary blob]\r\n	
The binary blob has the following format:			
<pre>#pragma pack(push,1) typedef struct reid_ble_packet_t { uint64_t time_ms; uint16_t vdd_mv; uint16_t fsr1; uint16_t fsr2; uint16_t fsr3; uint16_t fsr4; uint16_t fsr5; uint16_t fsr6; uint16_t fsr7; uint16_t fsr8; uint16_t fsr9; uint16_t fsr10; int16_t ntc1; int16_t ntc2; int16_t ntc3; int16_t ntc4; int16_t ntc5; uint16_t cap1s; uint16_t cap1n; uint16_t cap2s; uint16_t cap2n; uint16_t cap3s; uint16_t cap3n; uint16_t cap4s; uint16_t cap4n; } reid_ble_packet_t; #pragma pack(pop)</pre>			
Fresh data	F	;QF\r\n	Performs a single measurement and returns this measurement. This measurement is not considered in any of the summary generation. This allows you to “measure” at a rate faster than the normal rate for the device.
		;RF [binary blob]\r\n	

Command/Acknowledge Types

Command Type	Character	Example	Description
Reboot	R	;CR\r\n ;AR\r\n	Reboots the device, after a short timeout to allow for the response to be sent.
Erase All Measurements	E	;CE\r\n ;AE\r\n	Erases all of the stored measurements on the device. Note: because writing to and erasing flash requires the Bluetooth to be turned off, the acknowledge will be sent and a delay of 250ms observed, and then BLE disconnected and the command executed. Normal BLE will resume once the command has completed execution.
Set time	T	;CT 1643057462\r\n ;AT\r\n	Sets the internal timer to the amount specified; this can be used to set it to an epoch (like in this example), or could be used to set back to 0 (i.e. seconds since last sync).
Mark a record as synced	S	;CS 97\r\n ;AS\r\n	Marks the record with the specified index as synchronised, so that the controller may request the data and then only mark it once it has completed upload to the cloud. Note: because writing to and erasing flash requires the Bluetooth to be turned off, the acknowledge will be sent and a delay of 250ms observed, and then BLE disconnected and the command executed. Normal BLE will resume once the command has completed execution.
Mark multiple record as synced	M	;CM 97,102\r\n ;AM\r\n	Marks the records with indices in the specified range as synchronised, so that the controller may request the data and then only mark it once it has completed upload to the cloud. Note: because writing to and erasing flash requires the Bluetooth to be turned off, the acknowledge be sent and a delay of 250ms observed, and then BLE disconnected and the command executed. Normal BLE will resume once the command has completed execution.

Standard data download procedure

Controller	Peripheral	Description
;QT\r\n	;RT 1648097508\r\n	Confirm device time (incorrect time can indicate battery problems, or the device resetting)
;QP\r\n	;RP 97,102\r\n	Get the pending data indices to be downloaded
;QM 97,102\r\n	;RM [binary blob]\r\n ;RM [binary blob]\r\n ;RM [binary blob]\r\n ;RM [binary blob]\r\n ;RM [binary blob]\r\n	Download the data from the device, and check it for consistency. If any data looks corrupted, re-download to confirm.
;CS 97,102\r\n	;AS\r\n	Mark the data as synced. The peripheral will disconnect briefly while it does this.
;QP\r\n	;RP 4294967295,0\r\n	Check that the data was marked as synced, and that there is no more data to query.